

Problem 25.7

You are given

$$\frac{d}{dt}\bar{s}_{\overline{t}|} = (1.02)^{2t}.$$

Calculate δ .

Solution

For a continuous annuity accumulated to time t under a constant force of interest δ ,

$$\bar{s}_{\overline{t}|} = \frac{e^{\delta t} - 1}{\delta}.$$

Differentiate with respect to t :

$$\frac{d}{dt}\bar{s}_{\overline{t}|} = \frac{d}{dt} \left(\frac{e^{\delta t} - 1}{\delta} \right).$$

Since δ is constant,

$$\frac{d}{dt}\bar{s}_{\overline{t}|} = e^{\delta t}.$$

We are given

$$\frac{d}{dt}\bar{s}_{\overline{t}|} = (1.02)^{2t}.$$

Thus,

$$e^{\delta t} = (1.02)^{2t}.$$

Rewrite the right-hand side:

$$(1.02)^{2t} = \left((1.02)^2\right)^t = (1.0404)^t.$$

Therefore,

$$e^{\delta t} = (1.0404)^t.$$

Taking natural logarithms,

$$\delta t = t \ln(1.0404).$$

Canceling t ,

$$\delta = \ln(1.0404).$$

Equivalently,

$$\delta = 2 \ln(1.02).$$

Therefore,

$$\delta = 0.0396052.$$

Thus,

$$\boxed{\delta = 0.0396052}.$$